

CURRICULUM REVISION 2021 - PROGRAMME SCHEME													
STATE BOARD OF TECHNICAL EDUCATION, KERALA													
TEACHING AND EXAMINATION SCHEME FOR DIPLOMA PROGRAMME													
PROGRAM NAME : Electronics Engineering											PROGRAM CODE : EL		
PATTERN: FULL TIME													
DURATION OF PROGRAMME: 3 YEARS (6 SEMESTERS)													
SEMESTER: First Semester											SCHEME: Revision 2021		
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Humanities & Social Sciences	1001	Communication Skills in English	4	0	0	4	50	75	125	T	B	4
2.	Basic Science	1002	Mathematics I	5	0	0	5	50	75	125	T	B	5
3.	Basic Science	1003	Applied Physics I	3	0	0	3	50	75	125	T	B	3
4.	Basic Science	1004	Applied Chemistry	3	0	0	3	50	75	125	T	B	3
5.	Engineering Science	1005	Engineering Graphics	0	0	3	3	50	75	125	D	B	1.5
6.	Basic Science	2006	Applied Physics Lab	0	0	2	2	^	^	^	P	I	^
7.	Basic Science	1007	Applied Chemistry Lab	0	0	2	2	75	50	125	P	I	1
8.	Engineering Science	1008	Introduction to IT systems Lab	1	0	2	3	75	50	125	P	I	2
9.	Engineering Science	2009	Engineering Workshop Practice	0	0	3	3	^	^	^	P	I	^
10.	Humanities & Social Sciences	1009	Sports and Yoga	0	0	2	2	75	50	125	P	I	1
				16	0	14	30						20.5

PROGRAM NAME : Electronics Engineering										PROGRAM CODE : EL			
SEMESTER: Second Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Credits	
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		Evaluation Mode
L	T	P	Total										
1.	Basic Science	2002	Mathematics II	4	0	0	4	50	75	125	T	B	4
2.	Basic Science	2003	Applied Physics II	3	0	0	3	50	75	125	T	B	3
3.	Common Courses	2001	Environmental Science	3	0	0	3	50	75	125	T	I	0
4.	Engineering Science	2031	Fundamentals of Electrical & Electronics Engineering	3	0	0	3	50	75	125	T	B	3
5.	Engineering Science	2041	Basic Electronics	3	0	0	3	50	75	125	T	B	3
6.	Humanities & Social Sciences	2008	Communication Skills in English Lab	0	0	3	3	75	50	125	P	I	1.5
7.	Basic Science	2006	Applied Physics Lab	0	0	2	2	75	50	125	P	I	1
8.	Engineering Science	2039	Fundamentals of Electrical & Electronics Engineering Lab	0	0	3	3	75	50	125	P	I	0
9.	Engineering Science	2049	Electronics Tinkering Workshop	0	0	3	3	75	50	125	P	I	0
10.	Engineering Science	2009	Engineering Workshop Practice	0	0	3	3	75	50	125	P	I	1.5
				16	0	14	30						17
Summer Internship I to be conducted after Semester 2													
1.	Common Courses	3009	Internship I	3 Weeks				0	75	75	P	I	2

PROGRAM NAME : Electronics Engineering										PROGRAM CODE : EL			
SEMESTER: Third Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Programme core course	3041	Electric Circuits & Networks	2	1	0	3	50	75	125	T	B	3
2.	Programme core course	3042	Principles of Electronic Communication	2	1	0	3	50	75	125	T	B	3
3.	Programme core course	3043	Electronic Circuits	3	1	0	4	50	75	125	T	B	4
4.	Programme core course	3044	Digital Electronics	3	1	0	4	50	75	125	T	B	4
5.	Programme core course	3045	Fundamentals of C Programming	3	1	0	4	50	75	125	T	I	0
6.	Programme core course	3046	Principles of Electronic Communication Lab	0	0	3	3	75	50	125	P	B	1.5
7.	Programme core course	3047	Electronic Circuits Lab	0	0	3	3	75	50	125	P	B	1.5
8.	Programme core course	3048	Digital Electronics Lab	0	0	3	3	75	50	125	P	B	1.5
9.	Programme core course	3049	Fundamentals of C Programming Lab	0	0	3	3	75	50	125	P	B	1.5
				13	5	12	30						20

PROGRAM NAME : Electronics Engineering										PROGRAM CODE : EL			
SEMESTER: Fourth Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
L	T	P	Total										
1.	Programme core course	4041	Microcontroller and Applications	3	1	0	4	50	75	125	T	B	4
2.	Programme core course	4042	Electronic Measurements and Instrumentation	3	1	0	4	50	75	125	T	B	4
3.	Programme core course	4043	Linear Integrated Circuits	3	1	0	4	50	75	125	T	B	4
4.	Common Courses	4001	Community Skills in Indian knowledge system	2	0	0	2	50	75	125	T	I	0
5.	Programme core course	4046	Microcontroller and Applications Lab	0	0	3	3	75	50	125	P	B	1.5
6.	Programme core course	4047	Linear Integrated Circuits Lab	0	0	3	3	75	50	125	P	B	1.5
7.	Programme core course	4048	PCB and Prototyping Workshop	0	0	3	3	75	50	125	P	B	1.5
8.	Programme core course	4049	Python Programming Lab	0	1	2	3	75	50	125	P	I	0
9.	Minor project	4009	Minor Project	0	0	4	4	0	75	75	P	I	2
				11	4	15	30						18.5
Summer Internship II to be conducted after Semester 4													
1.	Common Courses	5009	Internship II	4 Weeks				0	75	75	P	I	3

PROGRAM NAME : Electronics Engineering										PROGRAM CODE : EL			
SEMESTER: Fifth Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
L	T	P	Total										
1.	Common Courses	5001	Industrial Management and Safety	3	1	0	4	50	75	125	T	I	0
2.	Programme core course	5041	Embedded Systems	3	1	0	4	50	75	125	T	B	4
3.	Programme core course	5042	Industrial Automation	3	1	0	4	50	75	125	T	B	4
4.	Program Elective course	5043A	Optical Communication and Networking	4	0	0	4	50	75	125	T	B	4
	Program Elective course	5043B	Microwave Devices and Radar	4	0	0	4	50	75	125	T	B	4
	Program Elective course	5043C	Advanced Microprocessors	4	0	0	4	50	75	125	T	B	4
	Program Elective course	5043D	Digital Communication	3	1	0	4	50	75	125	T	B	4
	Program Elective course	5043E	Telecommunication Networks	4	0	0	4	50	75	125	T	B	4
5.	Programme core course	5047	Embedded Systems Lab	0	0	3	3	75	50	125	P	B	1.5
6.	Programme core course	5048	Industrial Automation Lab	0	0	3	3	75	50	125	P	B	1.5
7.	Program Elective course	5049A	Microwave Engineering Lab	0	0	3	3	75	50	125	P	B	1.5
	Program Elective course	5049B	Digital Communication Lab	0	0	3	3	75	50	125	P	B	1.5
	Program Elective course	5049C	Telecommunication Networks Lab	0	0	3	3	75	50	125	P	B	1.5
8.	Seminar	5008	Seminar	0	0	2	2	0	75	75	P	I	1
9.	Major Project	6009	Major Project	0	0	3	3	0	0	0	P	I	0
				13	3	14	30						17.5

The Student has two options to complete the sixth semester and he/she shall select any one: 1. NORMAL PATHWAY 2. INTERNSHIP PATHWAY

1. NORMAL PATHWAY

PROGRAM NAME : Electronics Engineering

PROGRAM CODE : EL

SEMESTER: Sixth Semester

SCHEME: Revision 2021

Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Humanities & Social Sciences	6001	Entrepreneurship and Startup	3	1	0	4	50	75	125	T	I	4
2.	Program Elective course	6041A	Medical Electronics	4	0	0	4	50	75	125	T	B	4
	Program Elective course	6041B	Verilog HDL & Programmable Logic Devices	4	0	0	4	50	75	125	T	B	4
	Program Elective course	6041C	Consumer Electronics	4	0	0	4	50	75	125	T	B	4
3.	Open Elective course	6042A	Concepts of IoT	4	0	0	4	50	75	125	T	I	4
	Open Elective course	6042B	Contemporary Electronics	4	0	0	4	50	75	125	T	I	4
	Open Elective course	6042C	Introduction to Hybrid and Electric Vehicles	4	0	0	4	50	75	125	T	I	4
	Open Elective course	6042D	Introduction to Multimedia	4	0	0	4	50	75	125	T	I	4
4.	Common Courses	6002	Indian Constitution	2	0	0	2	50	75	125	T	I	0
5.	Programme core course	6047	Simulation Lab with Numerical software	0	0	3	3	75	50	125	P	I	1.5
6.	Programme core course	6048	Computer Hardware and Data Communication Lab	0	1	3	4	75	50	125	P	I	2.5
7.	Program Elective course	6049A	Medical Electronics Lab	0	0	3	3	75	50	125	P	B	1.5
	Program Elective course	6049B	Verilog HDL & PLD Lab	0	0	3	3	75	50	125	P	B	1.5
8.	Major Project	6009	Major Project	0	0	6	6	75	50	125	P	I	4
				13	2	15	30						21.5
													120

2. INTERNSHIP PATHWAY													
PROGRAM NAME : Electronics Engineering										PROGRAM CODE : EL			
SEMESTER: Sixth Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Humanities & Social Sciences	6001	Entrepreneurship and Startup	3	1	0	4	50	75	125	T	I	4
2.	Common Courses	6002	Indian Constitution	2	0	0	2	50	75	125	T	I	0
3.	Programme core course	6047	Simulation Lab with Numerical software	0	0	3	3	75	50	125	P	I	1.5
4.	Programme core course	6048	Computer Hardware and Data Communication Lab	0	1	3	4	75	50	125	P	I	2.5
5.	Internship	6007	Internship	4 - 6 months per semester				250	250	500	P	I	13.5
				5	2	6	13						21.5
Teaching Scheme : L-Lecture, T-Tutorial, P- Practical Examination Type : T-Theory, P-Practical, D-Drawing Examination Mode : B-Board Level, I-Institution Level ^ Carried over to next semester													